A Seminar Report on:

Crop Disease Detection

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ABSTRACT

Crop diseases are one of the major challenges in agriculture, as they affect plant health, reduce crop production, and cause financial losses to farmers. Early and accurate detection of crop diseases is essential for improving crop quality and agricultural productivity. Traditional methods of disease identification mainly depend on manual observation, which can be time-consuming and may require expert knowledge. To overcome these limitations, machine learning and deep learning techniques are being explored for the automatic detection and classification of crop diseases.

This project focuses on the study of crop disease detection using leaf images and modern artificial intelligence techniques. It examines different approaches, including image processing, Support Vector Machine (SVM), KNearest Neighbors (KNN), Convolutional Neural Network (CNN), ResNet, DenseNet, and EfficientNet. These techniques help identify disease-related patterns in leaf images and classify plants as healthy or affected by particular diseases. The study also considers transfer learning, mobile-based detection systems, offline solutions, and Explainable Artificial Intelligence (XAI) for improving accessibility and understanding model predictions.

The literature review analyzes twelve research papers to understand the methods used, their major findings, and their limitations. It highlights challenges such as limited datasets, changing environmental conditions, complex backgrounds, and similarities between disease symptoms. The study emphasizes the importance of testing detection models on diverse real-world images to improve their reliability. Overall, automated crop disease detection has the potential to support early diagnosis, reduce crop losses, and help farmers make better decisions, contributing to more efficient and sustainable agricultural practices.

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CHAPTER 1

INTRODUCTION

Agriculture is one of the most important sectors, as it provides food and supports the livelihood of millions of people. However, plant diseases are a major challenge in agriculture because they can affect crop quality, reduce production, and cause significant losses to farmers. Early identification of crop diseases is essential for taking timely action and maintaining healthy crops. Traditional disease detection methods generally depend on visual inspection by farmers or agricultural experts, which can be time-consuming and may not always provide accurate results.

With the development of Artificial Intelligence (AI), Machine Learning (ML), and Deep Learning (DL), automated crop disease detection has become a promising research area. These technologies can analyze images of plant leaves and identify disease-related patterns, such as changes in colour, spots, and unusual textures. Image processing techniques and machine learning algorithms, such as Support Vector Machine (SVM) and K-Nearest Neighbors (KNN), have been explored for disease classification. Advanced deep learning models, including Convolutional Neural Networks (CNN), ResNet, DenseNet, and EfficientNet, can automatically learn important features from images and classify different plant diseases.

Several researchers have developed crop disease detection systems using leaf image datasets. Some studies focus on improving classification accuracy through deep learning and transfer learning, while others explore mobile-based and offline systems to make the technology more accessible to farmers. Explainable Artificial Intelligence (XAI) is also being investigated to help users understand which parts of an image influence the model's prediction.

Despite these advancements, crop disease detection still faces challenges, including limited training data, different lighting conditions, complex backgrounds, and similarities between disease symptoms. Models trained on controlled images may not always perform equally well in real agricultural environments.
Therefore, studying existing research is important to understand the methods used, their performance, and their limitations.

This literature review examines twelve research papers related to crop disease detection. It discusses the techniques, algorithms, major findings, and limitations presented in these studies. The review also identifies research gaps that can guide the development of more accurate, reliable, and practical crop disease detection systems for agricultural applications.

CHAPTER 2

LITERATURE REVIEW

A literature review studies existing research on crop disease detection using image processing, machine learning, deep learning, and transfer learning. The following twelve papers discuss their techniques, achievements, and limitations.

1. Using Deep Learning for Image-Based Plant Disease Detection (2016)

Authors: Sharada Prasanna Mohanty, David P. Hughes, and Marcel Salathé.

Mohanty et al. (2016) proposed a deep learning-based approach for automatic plant disease detection using leaf images. The researchers used 54,306 images from the PlantVillage dataset, covering 14 crop species and 26 diseases, along with healthy leaves. A deep Convolutional Neural Network (CNN) was trained to identify crop types and disease categories. The model achieved 99.35% accuracy on the test dataset under controlled conditions. However, its accuracy decreased to 31.4% when tested on images collected from different online sources.The study demonstrated the potential of deep learning for automated crop disease detection and highlighted the importance of diverse training images.

2. Deep Learning for Image-Based Cassava Disease Detection (2017)

Authors: Amanda Ramcharan and colleagues.

Ramcharan et al. (2017) developed a deep learning approach for detecting diseases in cassava plants using leaf images captured in Tanzania. The researchers used transfer learning with a deep convolutional neural network to identify three diseases, two types of pest damage, and their relevant categoriesThe best model achieved an overall accuracy of 93% on images that were not used during training. The study demonstrated that image-based disease detection could provide an affordable and scalable solution for farmers, particularly in areas where agricultural experts are not easily available. However, the performance of the model depends on the quality and variety of training images.

3. A Comparative Study of Fine-Tuning Deep Learning Models for Plant Disease Identification (2018/2019)

Authors: Edna Chebet Too, Li Yujian, Sam Njuki, and Liu Yingchun.

Too et al. conducted a comparative study of different deep learning architectures for plant disease identification. The researchers evaluated VGG16, Inception V4, ResNet50, ResNet101, ResNet152, and DenseNet121 using the PlantVillage dataset. The dataset contained 38 classes of healthy and diseased leaves from 14 plant species. The models were fine-tuned using transfer learning to improve

their ability to classify plant diseases. Among the evaluated architectures, DenseNet121 achieved the highest testing accuracy of 99.75% and showed efficient learning with fewer parameters compared with several alternative models. The study established that selecting an appropriate pre-trained architecture can improve disease classification performance. However, the results depend on the dataset and may not directly represent performance under different field conditions.

4. A Robust Deep-Learning-Based Detector for Real-Time Tomato Plant Diseases and Pests Recognition (2017)

Authors: J. Fuentes and colleagues.

Fuentes et al. (2017) proposed a deep learning-based object detection system for identifying tomato plant diseases and pests. Unlike basic image classification systems, the proposed approach focused on locating disease symptoms and pests within plant images. The researchers investigated deep learning detection architectures, including Faster R-CNN, to recognize targets under challenging conditions such as changing illumination, overlapping leaves, and complex backgrounds. The study reported that the detector could identify disease-related regions in tomato plant images, making it useful for practical agricultural monitoring. The research highlighted the importance of object detection when disease symptoms must be located within a larger image. However, the system was focused on tomato-related categories, and applying it to other crops would require appropriate training data and model adaptation.

5. A Comprehensive Review on Detection of Plant Disease Using Machine Learning and Deep Learning Approaches (2022)

Authors: C. Jackulin and S. Murugavalli.

Jackulin and Murugavalli (2022) presented a comprehensive review of machine learning and deep learning techniques used for plant disease detection. The study examined methods such as logistic regression, image segmentation, traditional machine learning classifiers, artificial intelligence, and CNN-based approaches. It discussed how image processing and feature extraction can help identify disease symptoms from plant leaves. The authors compared the use and performance of machine learning and deep learning methods across existing research. The review highlighted the importance of automated disease identification for reducing crop losses and improving agricultural productivity. It also showed that deep learning, particularly CNNs, offers promising capabilities for image-based disease detection. However, segmentation methods and classification approaches may face challenges related to image noise, backgrounds, and variations in leaf appearance.

6. A Systematic Analysis of Machine Learning and Deep Learning Based Approaches for Identifying and Diagnosing Plant Diseases (2023)

Authors: Imtiaz Ahmed and Pramod Kumar Yadav.

Ahmed and Yadav (2023) studied different machine learning and deep learning approaches for the identification and diagnosis of plant diseases. The research used the PlantVillage dataset, which included 12 crop species and multiple disease categories. The approaches discussed and evaluated included Support Vector Machines (SVM), Gray-Level Co-occurrence Matrix (GLCM), Artificial Neural Networks (ANN), CNNs, and K-means clustering. Image processing and feature extraction were considered important steps in developing effective disease classification systems. The study emphasized that automated disease detection can reduce monitoring time and help farmers identify diseases at an early stage. It also demonstrated the importance of choosing suitable algorithms for different datasets. However, the performance of these methods may vary according to image quality, disease symptoms, and the characteristics of the dataset.

7. A Scalable Offline AI-Based Solution to Assist the Diseases and Plague Detection in Agriculture (2023)

Authors: Matias Urbieta, Martin Urbieta, Mauro Pereyra, Tomas Laborde, Guillermo Villarreal, and Mariana Del Pino.

Urbieta et al. (2023) introduced an AI-based solution for detecting plant diseases and pests using a mobile application that can operate without a continuous internet connection. The research focused on tomato diseases, particularly powdery mildew and Cladosporium. The system combined cloud-based model training with mobilebased disease detection using object detection techniques. It investigated models such as EfficientDet, Faster RCNN, and SSD MobileNet V2. The application was designed to identify disease symptoms directly from images captured in the field. The study highlighted the usefulness of offline detection for farmers working in areas with limited internet access. However, detection performance varied across models and disease categories, and field image diversity remained an important challenge.

8. Early Disease Detection in Plants Using CNN (2024)

Authors: Tejaswini, Priyanka Rastogi, Swayam Dua, Manikanta, and Vikas Dagar.

Tejaswini et al. (2024) proposed a CNN-based system for early detection of plant leaf diseases. The researchers used images of tomato, potato, and bell pepper leaves from the PlantVillage dataset. The model was developed to classify multiple healthy and diseased leaf categories using deep learning. The study reported a training accuracy of 86.21% for the CNN model, while the K-Nearest Neighbors (KNN) classifier achieved 82.5% accuracy. The results indicated that CNNs can learn complex visual patterns from leaf images and perform better than the compared KNN approach in this experiment. The research demonstrated the usefulness of CNNs for automated plant disease identification. However, further model optimization and validation on diverse field images would be useful for improving reliability.

9. Exploring Deep Learning and Explainable AI for Precise Rice Leaf Disease Detection and Classification (2025)

Authors: Tanjim Mahmud, Ms. Sharmin Akter, Tahmina Akter, Mohammad Shahadat Hossain, and Karl Andersson.

Mahmud et al. (2025) developed a deep learning-based approach for detecting and classifying rice leaf diseases. The study focused on bacterial leaf blight, brown spot, tungro, blast, and healthy leaves. Image preprocessing techniques such as Gaussian filtering and Contrast Limited Adaptive Histogram Equalization (CLAHE) were used to improve image quality. Canny and Prewitt edge detection methods were also applied to support feature extraction. The researchers integrated Explainable AI techniques, including Grad-CAM, Grad-CAM++, and LIME, to explain the predictions made by deep learning models. The study reported an accuracy of 99% and demonstrated how visual explanations can improve the transparency of disease classification. However, the limited number of disease samples was identified as a limitation, indicating the need for larger datasets and broader testing.

10. Deep Learning-Based Disease Detection in Potato and Mango Leaves: A Comparative Study of CNN, AlexNet, ResNet, and EfficientNet (2025/2026)

Authors: Utkarsh Mishra, Ansh Pandey, Logeswari G., and Tamilarasi K.

Mishra et al. investigated deep learning-based disease detection in potato and mango leaves. The researchers used two public datasets: the PlantVillage Potato Leaf Disease dataset containing 2,152 images and the Kaggle Mango Leaf Disease dataset containing 4,000 images. The images were preprocessed and augmented before being divided into training and testing sets. Four architectures—CNN, AlexNet, ResNet, and EfficientNet— were evaluated for multiclass disease classification. EfficientNet achieved the best reported validation accuracy of 97.8%, while ResNet achieved 96.7% validation accuracy. The results showed that the choice of deep learning architecture can significantly affect classification performance. However, visually similar disease symptoms and variations in realworld image conditions remain important challenges for practical deployment.

11. Evaluation of Deep Learning Models Using Explainable AI for Rice Leaf Disease Detection (2025)

Authors: Hari Kishan Kondaveeti and Chinna Gopi Simhadri.

This research focuses on detecting rice leaf diseases using deep learning models and Explainable Artificial

Intelligence (XAI). The researchers evaluated eight pre-trained models, including ResNet50, DenseNet201, EfficientNetB0, VGG16, and AlexNet. LIME was used to explain the features influencing model predictions. ResNet50 achieved the highest classification accuracy of 99.13%. The study highlights that evaluating both accuracy and model explanations can improve the reliability of

disease detection systems. However, some models may focus on irrelevant image features, affecting their performance in real-world conditions.

12. Deep Learning for Rice Leaf Disease Detection: A Systematic Literature Review on Emerging Trends, Methodologies and Techniques (2025)

Authors: Check the original paper's author list before adding it to your report.

This research presents a systematic review of 82 research papers on rice leaf disease detection published since 2017. It examines different techniques, including image processing, machine learning, deep learning, transfer learning, ensemble learning, and hybrid approaches. The review explains how deep learning models can automatically learn patterns from leaf images and classify plant diseases. It also discusses challenges such as limited datasets, similar disease symptoms, and changing environmental conditions. The study highlights the need for more accurate and reliable models that can perform effectively in real agricultural environments.

Table No. 2.1: Literature Review

Sr. No.	Approach Type	Technologies Used	Dataset	Key Objective	Limitations
1	Deep Learning	CNN, AlexNet, GoogLeNet	PlantVillage – 54,306 images	Detect and classify plant diseases automatically.	Controlled images; poor generalization to field conditions.
2	Deep Learning	Deep CNN, ECOC classifier	tomato leaf images	Identify and classify tomato leaf diseases.	Limited to selected tomato diseases; field testing is limited.
3	Image Segmentation + DL	CNN, image segmentation	Segmented plant leaf images	Improve disease classification by removing unnecessary background information.	Segmentation errors may affect classification accuracy.
4	Image Processing	Segmentation, feature extraction, classification	Plant leaf images	Identify diseased portions of leaves and classify plant diseases.	Performance depends on image quality and selected features
5	Hybrid Deep Learning	Convolutional Autoencoder (CAE), CNN	PlantVillage peach leaf images	Build a compact model to detect bacterial spot disease.	Limited crop and disease coverage.
6	Object Detection	Improved YOLOv4.	Apple disease images from orchard-like conditions.	Detect and locate diseases in real time.	Requires annotated images and computational resources.
7	Transfer Learning	DenseNet-121, ResNet-50, VGG-16, Inception V4	PlantVillage – 54,305 images, 38 classes.	Compare pretrained models for multiple crop disease classification.	Dataset may not represent real field conditions.

8	Literature Review	CNN, DCNN, transfer learning, data augmentation	PlantVillage, Kaggle, and other research datasets	Review CNN-based plant leaf disease detection techniques.	Dataset diversity and real-world validation remain challenging.
9	CNN-Based Classification	Custom 5-layer CNN, VGG-16, VGG-19, Xception, ResNet	Rice leaf images – 1,600 images	Detect and classify rice leaf diseases for early diagnosis.	Relatively small dataset may limit generalization.
10	Deep Learning + Explainable AI	Grad-CAM, Grad-CAM++, LIME, Captum	Rice leaf dataset – 7,420 images	Classify rice diseases and explain predictions visually.	Insufficient disease samples and limited generalization testing.
11	Hyperspectral Image Analysis	VNIR/SWIR imaging, machine learning, image processing	Capsicum plants infected with Tomato Spotted Wilt Virus	Detect infection using spectral information from plant leaves.	Requires specialized imaging equipment and processing.
12	Transfer Learning / Deep Learning	Pretrained CNN models and ensemble learning.	Rice leaf disease images	Improve rice disease classification by comparing and combining predictions from pretrained models.	Performance depends on dataset size, image diversity, and computational resources.

CHAPTER 3

INFORMATION / CHAPTERS

3.1 Meaning of Crop Disease Detection

Crop Disease Detection refers to the process of identifying diseases in plants by examining their visible symptoms, such as spots, discoloration, yellowing, and changes in leaf texture. Traditional disease identification mainly depends on manual observation by farmers and agricultural experts. However, this process can be time-consuming and may require specialized knowledge.

With advancements in Artificial Intelligence (AI), Machine Learning (ML), Deep Learning (DL), and image processing, crop diseases can be detected automatically using digital images. The system analyzes images of crop leaves and identifies patterns associated with different plant diseases.

A Crop Disease Detection System generally accepts an image of a plant leaf as input, processes the image, extracts useful features, and classifies it using a trained machine learning or deep learning model. The final result indicates whether the plant appears healthy or affected by a particular disease.

This technology can support early disease identification, help farmers make informed decisions, and contribute to improved crop management. However, the accuracy of the system depends on image quality, dataset diversity, and the performance of the trained model.

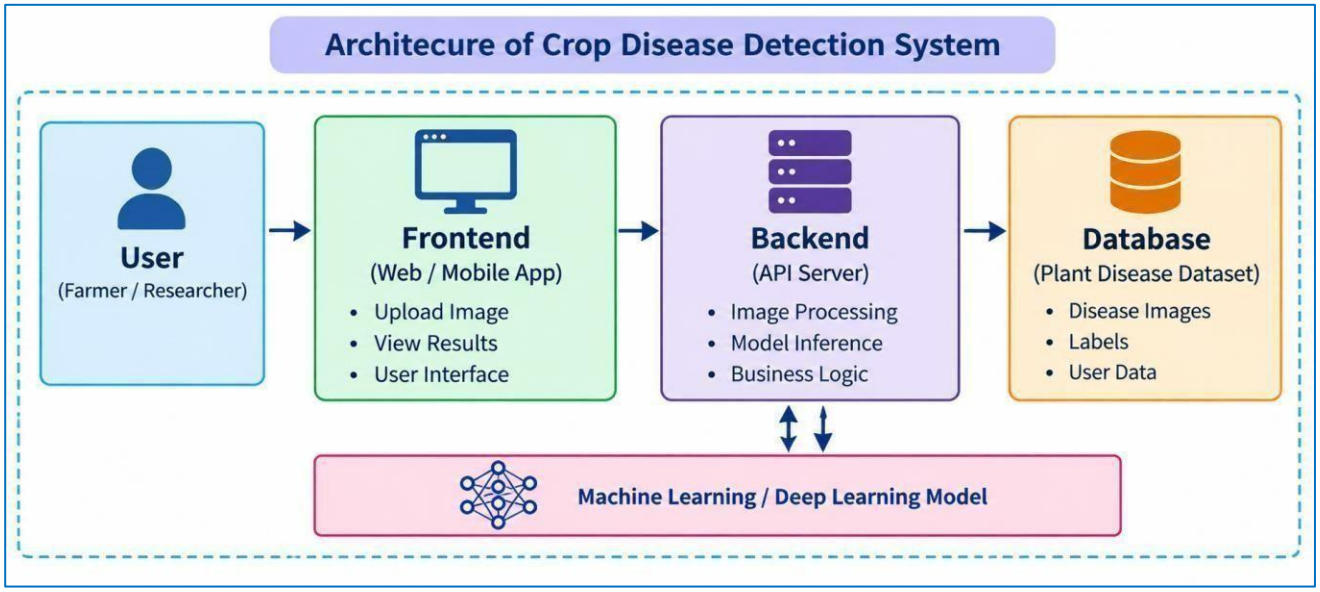


Fig no: 3.1 Architecture of Crop Disease Detection System

3.2 Common AI Techniques Used in Crop Disease Detection

1. **Machine Learning (ML):** Machine learning algorithms learn patterns from plant images and use them to classify healthy and diseased crops.
2. **Deep Learning (DL):** Deep learning models automatically learn complex visual patterns from large collections of plant images.
3. **Convolutional Neural Network (CNN):** CNN is widely used for image classification. It can identify features such as leaf spots, color changes, and disease patterns.
4. **Image Processing:** Image processing techniques improve image quality through resizing, noise removal, and normalization before classification.
5. **Transfer Learning:** Transfer learning uses a pretrained model and adapts it to a plant disease detection task. It can reduce the amount of training required.
6. **Data Augmentation:** Data augmentation creates modified training images through operations such as rotation, flipping, and zooming to improve model generalization.

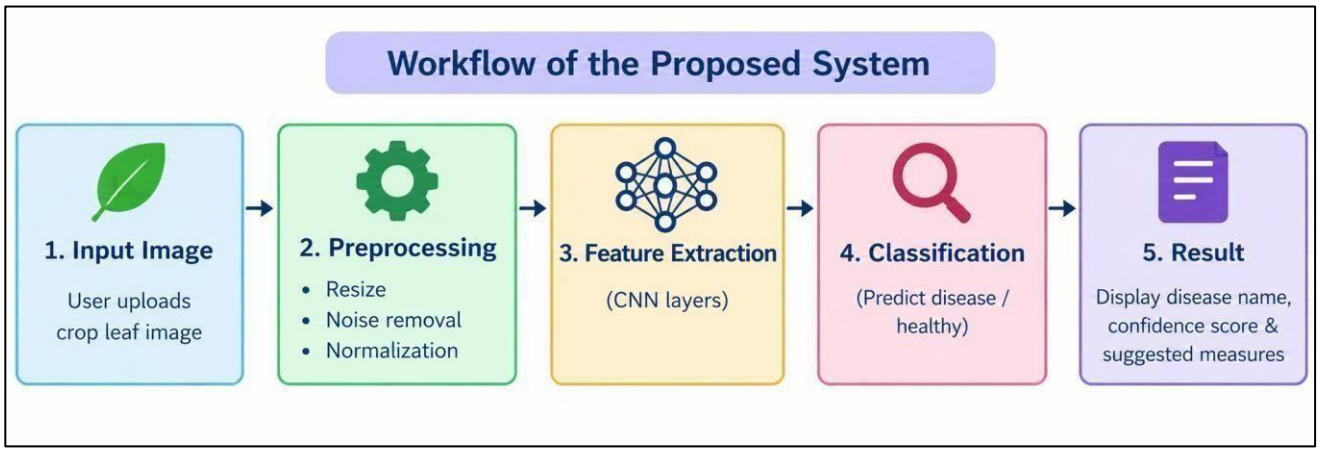


Fig no: 3.2 Workflow of the Proposed System

3.3 Applications of Crop Disease Detection

1. **Early Disease Identification:** Helps identify possible crop diseases before they spread extensively.
2. **Automated Leaf Analysis:** Examines leaf images to detect visible symptoms of plant diseases.
3. **Disease Classification:** Classifies plant images into healthy or diseased categories and may identify a specific disease.
4. **Smart Agriculture:** Supports the use of AI-based technologies in modern agricultural practices.
5. **Farmer Assistance:** Provides preliminary disease information that can help farmers decide whether expert advice is required.
6. **Crop Health Monitoring:** Supports the observation and assessment of plant health over time.
7. **Agricultural Research:** Helps researchers study disease symptoms and evaluate automated classification techniques.

8. **Disease Management Support:** Can provide general information about preventive measures and appropriate next steps for managing crop diseases.

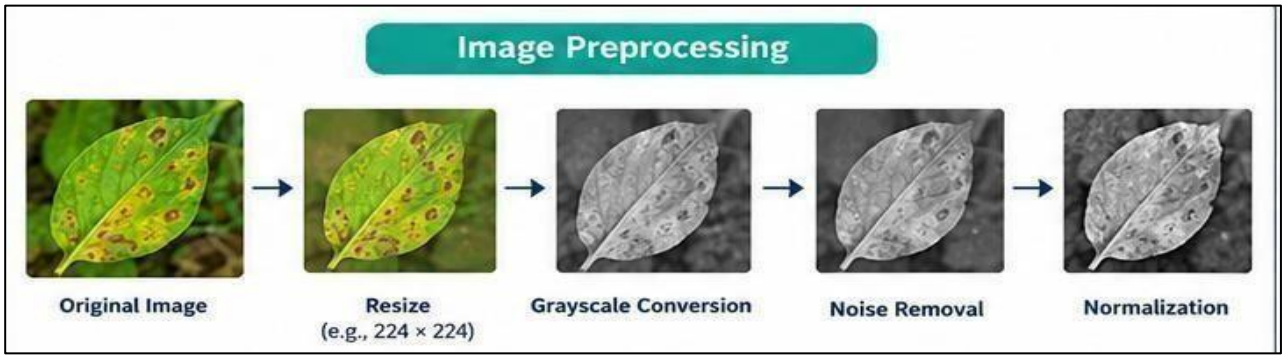


Fig no: 3.3 Image Preprocessing

3.4 Typical Workflow of Crop Disease Detection

Step 1: Image Collection: Images of healthy and diseased crop leaves are collected from suitable datasets or captured using a camera.

Step 2: Image Upload: The user uploads a crop leaf image through the system interface.

Step 3: Image Preprocessing: The image is resized to the required dimensions, and suitable preprocessing techniques are applied to prepare it for analysis.

Step 4: Feature Extraction: The system identifies useful visual patterns, such as leaf spots, texture, shape, and color variations. In a CNN, these features are learned automatically during training.

Step 5: Disease Classification: The trained machine learning or deep learning model analyzes the image and predicts the most likely disease category.

Step 6: Result Generation : The system displays the predicted disease or healthy status. Depending on the implementation, it may also show a confidence score and basic preventive information.

Step 7: Evaluation and Verification: The model's performance is evaluated using suitable test images and metrics such as accuracy, precision, recall, and F1-score. Important predictions should be verified by an agricultural expert when necessary.

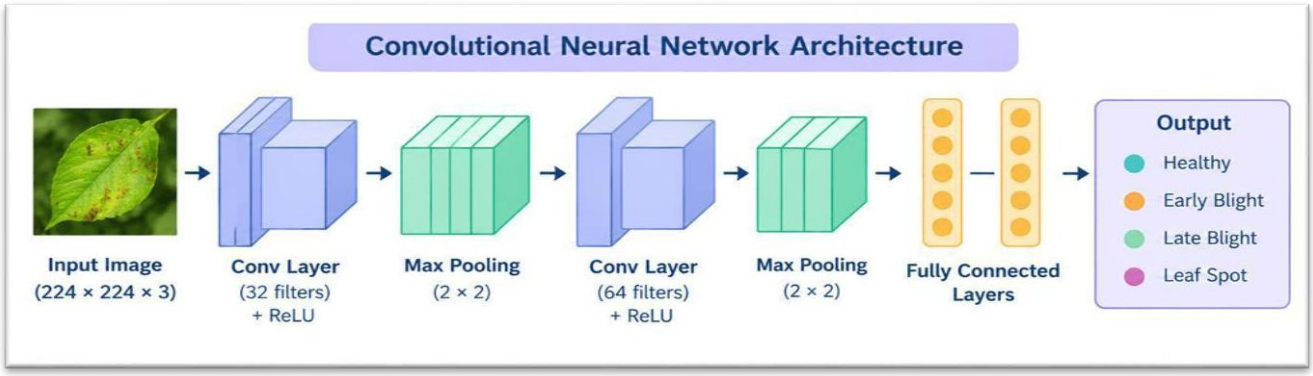


Fig no: 3.4 Convolutional Neural Network Architecture

3.5 Evaluation Metrics

The evaluation metrics are used to measure the performance and effectiveness of the Crop Disease Detection System. These metrics help determine how accurately the system identifies healthy and diseased crop leaves.

A combination of different evaluation metrics is necessary to understand the overall performance of the model.

1. **Accuracy:** Accuracy measures the percentage of crop images that are correctly classified by the model as healthy or diseased.
2. **Precision:** Precision measures how many images predicted as belonging to a particular disease category are actually from that category. It helps evaluate the reliability of disease predictions.
3. **Recall:** Recall measures how many actual diseased crop images are correctly identified by the system. A high recall is important to reduce missed disease cases.
4. **F1-Score:** The F1-score represents the balance between precision and recall. It is useful when evaluating the model on datasets with different numbers of images in each disease category.
5. **Disease Detection Rate:** This metric measures the proportion of diseased crop images correctly detected by the system. It helps assess the model's ability to identify affected plants.
6. **False-Positive Rate:** The false-positive rate measures how often healthy crop images are incorrectly classified as diseased. A lower false-positive rate helps reduce unnecessary concern and avoid inappropriate treatment decisions.
7. **Processing Time:** Processing time measures how long the system takes to analyze a crop image and generate a prediction. Lower processing time can improve the user experience.
8. **Model Generalization:** This evaluates how well the trained model performs on new crop images that were not included in the training dataset. It is important for assessing whether the system may be useful in real agricultural environments.

These metrics help researchers and developers assess the performance of the Crop Disease Detection System and identify areas requiring improvement. No single metric is sufficient to evaluate the

complete system. Therefore, accuracy, precision, recall, F1-score, false-positive rate, and processing time should be considered together.Overall, the use of suitable evaluation metrics supports reliable model assessment and helps improve the effectiveness of automated crop disease detection. The actual performance values should be reported only after testing the trained model on an appropriate test dataset.

3.6 Final Disease Detection Result

The final disease detection result is the output generated by the Crop Disease Detection System after analyzing the uploaded crop leaf image. The trained deep learning model processes the image and predicts whether the crop is healthy or affected by a particular disease.

The result may display the predicted disease name, confidence score, crop information, and suggested preventive measures. This information helps users understand the model's prediction and take appropriate next steps for crop management.

The final result is presented through a simple and user-friendly interface so that farmers and other users can easily understand the detected disease. However, the prediction should be treated as preliminary information and verified by an agricultural expert when necessary.



Fig no: 3.5 Final Disease Detection Result

CHARPTER 4

ADVANTAGES AND DISADVANTAGES

4.1 Advantages:

1. Early disease detection helps farmers take timely action.
2. Machine learning and deep learning techniques can identify diseases from leaf images.
3. Automated detection saves time compared to manual inspection.
4. Early identification helps reduce crop losses.
5. Automated systems can reduce the need for repeated manual inspection.
6. Advanced models can classify different types of plant diseases.
7. Mobile-based and offline systems can make disease detection more accessible.
8. Detection results help farmers make better crop management decisions.

4.2 Disadvantages:

1. Detection accuracy depends on the quality and quantity of training data.
2. Different lighting conditions and backgrounds may affect performance.
3. Diseases with similar symptoms can be difficult to distinguish.
4. Deep learning models may require powerful hardware and processing time.
5. Models trained on controlled datasets may not perform equally well in real fields.
6. Training advanced models may require technical knowledge.
7. Limited images of certain diseases can reduce classification accuracy.
8. Testing on diverse real-world images is necessary for reliable results.

CHAPTER 5

CONCLUSION

Crop disease detection using machine learning and deep learning has become an important approach for identifying plant diseases accurately and at an early stage. The reviewed research papers demonstrate that different techniques, such as image processing, Support Vector Machine (SVM), KNearest Neighbors (KNN), Convolutional Neural Network (CNN), DenseNet, ResNet, and EfficientNet, can be used to classify plant diseases based on leaf images. Among these techniques, deep learning models have shown promising results in automatic disease detection and classification. Transfer learning can also improve model performance while reducing the effort required to train models from scratch.

The studies also highlight the importance of mobile-based and offline disease detection systems, which can help farmers identify plant diseases without depending entirely on internet connectivity. In addition, Explainable Artificial Intelligence (XAI) techniques help users understand which parts of a leaf image influence the model's prediction. These developments show that automated crop disease detection can support farmers in making better decisions and taking timely action to protect their crops.

However, some challenges still exist, including limited training data, changing lighting conditions, complex backgrounds, and similarities between disease symptoms. A model that performs well on a controlled dataset may not always provide the same accuracy in real agricultural environments. Therefore, further research is needed to improve model reliability, test performance on diverse realworld images, and develop systems that are easy for farmers to use.

In conclusion, machine learning and deep learning offer promising solutions for crop disease detection. An accurate and reliable system can help identify diseases at an early stage, reduce crop losses, support timely treatment, and improve agricultural productivity. By combining effective algorithms with diverse datasets and practical testing, crop disease detection systems can contribute to smarter, more efficient, and sustainable agricultural practices.

CHAPTER 6

PROJECT INITIATION

6.1 Description of Problem Statement

Agriculture plays an important role in food production and economic development. However, crop diseases are a major problem that affects crop growth, quality, and productivity. Diseases caused by fungi, bacteria, viruses, and other harmful organisms can damage plant leaves and other parts of crops, resulting in reduced agricultural production and financial losses for farmers.

Traditional crop disease identification mainly depends on manual observation and the knowledge of farmers or agricultural experts. This process can be time-consuming and may not always provide accurate identification, especially when different diseases show similar symptoms. In addition, agricultural experts may not be easily available in every rural area.

Therefore, there is a need for an automated system that can help identify crop diseases quickly and efficiently. Artificial Intelligence, Machine Learning, Deep Learning, and image processing techniques can be used to analyze crop leaf images and recognize visible disease patterns.

The proposed Crop Disease Detection System allows users to upload an image of a crop leaf. The system processes the image and uses a trained classification model to predict whether the crop appears healthy or affected by a particular disease. The result may also include the predicted disease name, confidence score, and basic preventive information.

This project aims to study how automated image-based disease detection can support early identification and help farmers make informed decisions about crop health.

6.2 Scope and Objectives

Scope

- Develop a proposed crop disease detection system using plant leaf images.
- Study image processing techniques such as resizing, noise removal, and normalization.
- Explore machine learning and deep learning techniques for crop disease classification.
- Use a suitable plant disease image dataset for model training and evaluation.
- Display the predicted disease category and healthy status through a simple user interface.
- Explore the possibility of providing confidence scores and basic preventive information.
- Evaluate the model using suitable metrics such as accuracy, precision, recall, and F1-score.
- Limit the project to educational and research purposes.

Objectives

- To understand the concept of crop disease detection using Artificial Intelligence.
- To study the applications of machine learning and deep learning in agriculture.
- To identify common crop diseases through the analysis of leaf images.
- To understand the working of Convolutional Neural Networks (CNNs) for image classification.
- To study image preprocessing and feature extraction techniques used in disease detection.
- To explore suitable datasets for training and testing crop disease detection models.
- To evaluate the effectiveness of the proposed system using appropriate evaluation metrics.
- To identify the advantages, limitations, and challenges of automated crop disease detection.
- To explore future improvements in AI-based agricultural disease identification.

6.3 Stakeholders

- **Project Guide:** Provides guidance, monitors progress, and reviews the project work.
- **Students / Researchers:** Collect information, study research papers, and prepare the report on crop disease detection.
- **Farmers:** Use the system to obtain preliminary information about possible crop diseases.
- **Agricultural Experts:** Provide expert knowledge about crop diseases and help verify disease identification.
- **AI / ML Developers:** Develop and evaluate machine learning or deep learning models for crop disease classification.
- **Dataset Providers:** Provide plant leaf images and disease labels required for model training and testing.
- **System Developers:** Develop the user interface and integrate the trained model into the proposed system.
- **Educational Institutions:** Support research, learning, and experimentation in AI-based agricultural applications.

6.4 Seminar Discussion Plan

- Introduction to Crop Disease Detection and its importance in agriculture.
- Meaning and importance of Artificial Intelligence in agriculture.
- Common crop diseases and their visible symptoms.
- Common AI techniques, including Machine Learning, Deep Learning, and Convolutional Neural Networks.
- Applications of image processing in crop disease detection.

- Typical workflow of the proposed Crop Disease Detection System.
- Architecture of the system, including frontend, backend, database, and classification model.
- Image preprocessing and feature extraction techniques.
- Review and comparison of research papers related to crop disease detection.
- Evaluation metrics used to measure disease classification performance.
- Advantages and disadvantages of automated crop disease detection.
- Challenges, limitations, and future scope of AI-based agricultural systems.

6.5 Platform / Equipment / Resources Identification

1.Platform:

The Crop Disease Detection system can be developed using the Python programming language on platforms such as Google Colab, Jupyter Notebook, or VS Code. TensorFlow and Keras are used to build and train deep learning models such as Convolutional Neural Networks (CNN), while OpenCV is used for image processing. NumPy and Matplotlib help in data handling and visualization. The PlantVillage dataset, available on Kaggle, can be used to train and test the model for detecting and classifying crop diseases from leaf images.

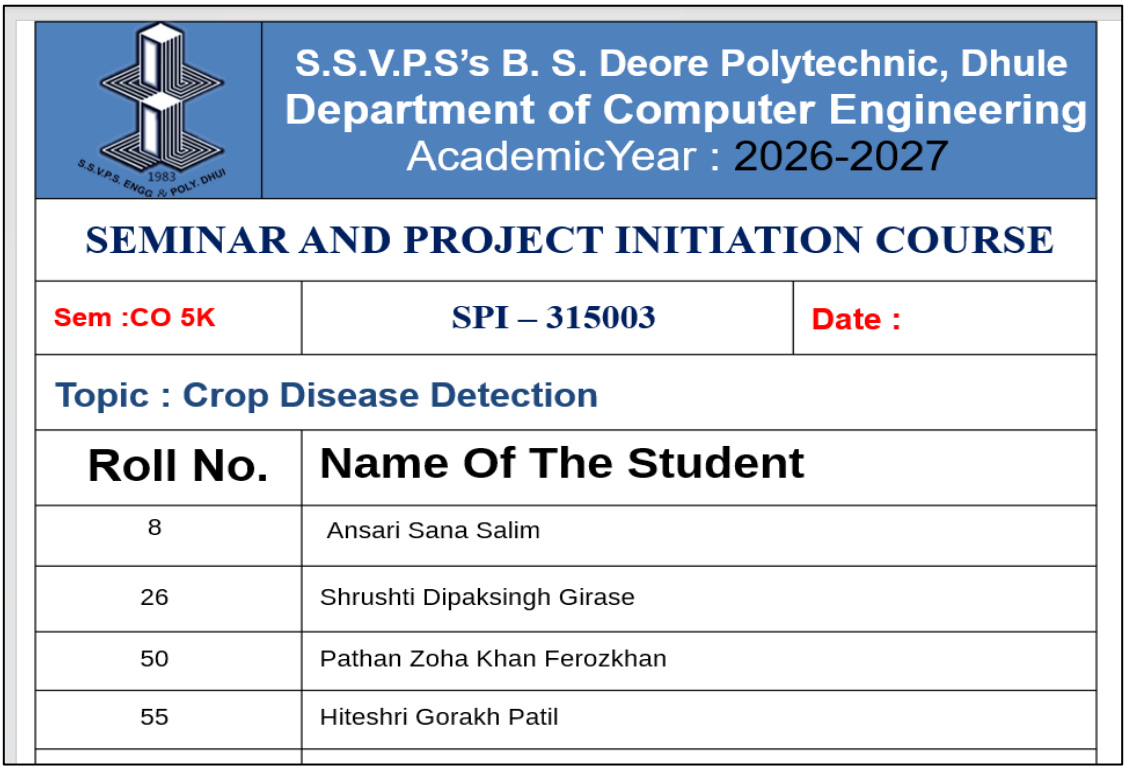
2.Equipment:

- Computer/Laptop: To develop and run the crop disease detection system.
- Smartphone/Digital Camera: To capture images of crop leaves.
- GPU (Optional): To speed up deep learning model training.
- Internet Connection: To access datasets and cloud platforms.
- Storage Device: To store crop images, datasets, and trained models.
- RAM and Processor: To process images and run the detection model efficiently.
- Specialized Camera (Optional): To capture detailed images for advanced disease analysis.

3.Resources:

Sr.No.	Resource	Purpose
1.	Hardware – Laptop/Computer	To develop and test the system.
2.	Software – Python	To develop the disease detection model.
3.	Deep Learning – CNN	To identify and classify crop diseases.
4.	Dataset – PlantVillage	To train and test the model using leaf images.
5.	Libraries – TensorFlow, OpenCV	To process images and build the model.
6.	IDE – VS Code / Google Colab	To write and execute code.
7.	Storage – MySQL / Local Storage	To store required data and results, if needed.

SEMINAR PRESENTATION



S.S.V.P.S's B. S. Deore Polytechnic, Dhule
Department of Computer Engineering
AcademicYear : 2026-2027

SEMINAR AND PROJECT INITIATION COURSE

Sem :CO 5K	SPI – 315003	Date :

Topic : Crop Disease Detection

Roll No.	Name Of The Student
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26	Shrushti Dipaksingh Girase
50	Pathan Zoha Khan Ferozkhan
55	Hiteshri Gorakh Patil

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Abstract

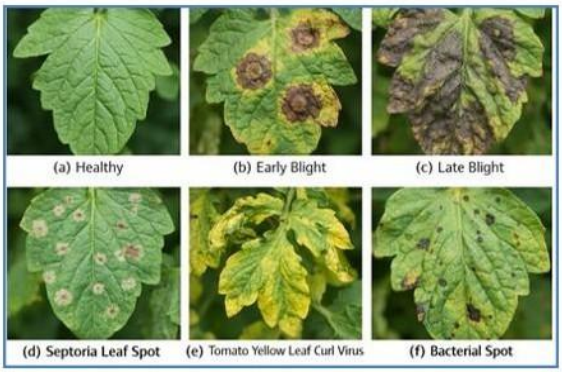
Crop Disease Detection is a technology-based system that identifies plant diseases using crop leaf images. It uses image processing and deep learning techniques, such as Convolutional Neural Networks (CNN), to classify leaves as healthy or diseased. The system aims to provide fast and easy disease detection, helping farmers take timely preventive measures, reduce crop losses, and improve agricultural productivity.

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1. Introduction

Agriculture is an importantpart of our lives, butcrop diseases can reduce crop quality and production. Traditional methods of disease detection require manual inspection and expert knowledge, which can be time-consuming.

Our project, Crop Disease Detection, uses image processing and deep learning techniques to identify diseases from crop leaf images. It aims to detect diseases at an early stage, help farmers take timely action, reduce crop losses, and improve agricultural productivity through modern technology.



2.Literatutre Survey

Sr. No	Approach Types	Technologies Used	Dataset	Key Objectives	Limitations
1	Deep Learning	CNN, AlexNet, GoogLeNet	PlantVillage – 54,306 images	Detect and classify plant diseases automatically.	Controlled images; poor generalization to field conditions.
2	Deep Learning	Deep CNN, ECOC classifier	PlantVillage – tomato leaf images	Identify and classify tomato leaf diseases.	Limited to selected tomato diseases; field testing is limited.
3	Image Segmentation + DL	CNN, image segmentation	Segmented plant leaf images	Improve disease classification by removing unnecessary background information.	Segmentation errors may affect classification accuracy.

2.Literatutre Survey

Sr. No.	Approach Type	Technology Used	Dataset	Key Objective	Limitations
4	Image Processing	Segmentation, feature extraction, classification	Plant leaf images	Identify diseased portions of leaves and classify plant diseases.	Performance depends on image quality and selected features.
5	Hybrid Deep Learning	Convolutional Autoencoder (CAE), CNN	PlantVillage peach leaf images	Build a compact model to detect bacterial spot disease.	Limited crop and disease coverage.
6	Object Detection	Improved YOLOv4	Apple disease images from orchard-like conditions	Detect and locate disease spots in real time.	Requires annotated images and computational resources.

2.Literatutre Survey

Sr. No.	Approach Type	Technology Used	Dataset	Key Objective	Limitations
7	Transfer Learning	DenseNet-121, ResNet-50, VGG-16, Inception V4	PlantVillage – 54,305 images, 38 classes	Compare pretrained models for multiple crop disease classes.	Controlled dataset may not represent real field conditions.
8	Literature Review	CNN, DCNN, transfer learning, data augmentation	PlantVillage, Kaggle, and other research datasets	Review CNN-based plant leaf disease detection techniques.	Dataset diversity and real-world validation remain challenging.
9	CNN-Based Classification	Custom 5-layer CNN, VGG-16, VGG-19, Xception, ResNet	Rice leaf images – 1,600 images	Detect and classify rice leaf diseases for early diagnosis.	Relatively small dataset may limit generalization.

2.Literatutre Survey

Sr. No.	Approach Type	Technology Used	Dataset	Key Objective	Limitations
10	Deep Learning + Explainable AI	Grad-CAM, Grad-CAM++, LIME, Captum	Rice leaf dataset – 7,420 images	Classify rice diseases and explain predictions visually.	Insufficient disease samples and limited generalization testing.
11	Hyperspectral Image Analysis	VNIR/SWIR imaging, machine learning, image processing	Capsicum plants infected with Tomato Spotted Wilt Virus	Detect infection using spectral information from plant leaves.	Requires specialized imaging equipment and processing.
12	Transfer Learning / Deep Learning	Pretrained CNN models and ensemble learning	Rice leaf disease images	Improve rice disease classification by comparing and combining pretrained models.	Performance depends on dataset size, image diversity, and computational resources.

3. Information To Seminar Topic

Crop Disease Detection is a technology-based approachusedto identifydiseasesin plants by analysing images of their leaves. Crop diseases can affect plant growth, reduce crop quality, and cause agricultural losses.

This system uses image processing, machine learning, and deep learning techniques such as Convolutional Neural Networks (CNN) to classify crop leaves as healthy or diseased. It can help farmers identify possible diseases early and take suitable preventive measures.

The main aim of this project is to make disease detection faster, easier, and more efficient. In the future, this technology can be improved to support more crops and disease types through web and mobile applications.

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1. Technology Used: Artificial Intelligence (AI), Machine Learning (ML), Convolutional Neural Networks (CNN), Python, OpenCV, and TensorFlow are used to analyze crop leaf images and identify possible diseases.

2. Approach: The system collects or accepts crop leaf images, preprocesses them, and analyzes them using a trained CNN model. The model predicts whether the crop is healthy or affected by a particular disease and displays the result.

3. Main Purpose: The main purpose is to detect crop diseases at an early stage using image processing and AI techniques, helping farmers make better decisions about crop health.

4. Key Concepts: The key concepts include image preprocessing, feature extraction, deep learning, image classification, model training, and performance evaluation.

5. Benefits: It supports early disease identification, saves time, reduces manual effort, and helps farmers manage crop health more effectively. It can also provide preliminary disease information quickly.

6. Applications: Crop Disease Detection can be used in smart farming, greenhouses, plant nurseries, agricultural research, farmer advisory systems, and mobile-based agricultural applications.

4. Advantages / Disadvantages

Advantages

:

- Detects crop diseases at an early stage.
- Saves time and manual effort.
- Helps reduce crop losses.
- Provides fast disease identification.
- Supports smart and modern farming.

Disadvantages

:

- Requires a good-quality image dataset.
- Poor images may give incorrect results.
- May detect only trained disease types.
- Requires computing resources for model training.
- Accuracy depends on the quality of the dataset.

5. Conclusion

Crop Disease Detection is a useful application of modern technology in agriculture. By using image processing, Machine Learning, and Deep Learning, the system can help identify possible crop diseases from leaf images. Early detection allows farmers to take timely action and manage crops more effectively. Although challenges such as image quality and model accuracy remain, this technology has the potential to reduce crop losses and improve agricultural productivity.

6. Project Initiation

- The project initiation phase focuses on identifying the problem and planning the proposed Crop Disease Detection System.

1. **Problem Identification:** Identified the need for early detection of crop diseases to reduce crop losses.
2. **Project Objectives:** Aim to detect and classify crop diseases using images of plant leaves.
3. **Literature Review:** Studied existing research papers on AI-based crop disease detection.
4. **Research Gap:** Examined the limitations of existing approaches, such as performance under real-field conditions and dependence on internet connectivity.
5. **Technology Selection:** Considered Artificial Intelligence, Deep Learning, and Image Processing for disease detection.
6. **Feasibility Study:** Evaluated the technical requirements, available resources, and feasibility of the proposed system.
7. **Project Scope:** Defined the basic functions and expected outcomes of the system.

6.1 Description of Problem Statement

Crop Disease Detection Using Deep Learning is an AI-based project designed to identify plant diseases from leaf images. It uses deep learning techniques, such as Convolutional Neural Networks (CNN), to classify crop leaves and detect possible diseases. This system can help farmers identify diseases early and take appropriate action. The main aim of the project is to support smart agriculture, reduce crop losses, and improve crop health.

6.2 Scope and Objectives.

- Detect crop diseases using plant leaf images.
- Identify healthy and diseased leaves.
- Support early detection of crop diseases.
- Use AI and Machine Learning for disease classification.
- Reduce the time and effort required for disease identification.
- Help farmers take timely preventive measures and reduce crop losses.
- Improve crop monitoring and agricultural productivity through modern technology.



6.3 Stake Holder

- Farmers
- Agricultural experts
- Researchers
- Students and developers
- Agricultural organizations

6.5 Platform/ Equipment/ Resources

- **Hardware:** Computer or laptop, camera or smartphone.
- **Software:** Python, Jupyter Notebook or Google Colab.
- **Libraries:** OpenCV, TensorFlow or Keras, NumPy.
- **Dataset:** PlantVillage or other suitable crop leaf datasets.

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Thank you

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SEMINAR AND PROJECT INITIATION ASSESSMENT / EVALUATION

Sr. No	Enrollment Number	Seat Number	Name of student
1	24610960175	388495	Ansari Sana Salim
2	24610960199	388513	Girase Shrushti Dipaksing
3	24610960229	388537	Pathan Zoha Khan Ferozkhan
4	24610960234	388542	Patil Hiteshri Gorakh

Formative Assessment

Sr. No.	Criteria	Weightage	Examination Seat Number			
			388495	388513	388537	388542
1	Selection Topic / Theme of seminar	5				
2	Literature review and data presentation	5				
3	Quality of Preparation and innovativeness	5				
4	Q-A handling	5				
5	Time Management	5				
6	Seminar Presentation report	10				
7	Selection of Theme of Problem Statement and its innovativeness	5				
8	Stages of development of Action plan	5				
9	Prototyping	5				
	Total	**50**				
	Scaled to 25					

Sign :

Name : **Shraddha S Hajari**

(Course Expert/s)